Proficient in Ansys (80 Hours)

S.No	Session	Topics	Type	Duration (Hrs)
1	Session 1&2	**INTRODUCTION TO FINITE ELEMENT ANALYSIS** **CAE:** Introduction to CAE, CAE Tools, CAE in Automotive industry, CAE in Aerospace & Defence industry, CAE in Construction Computational Fluid Dynamics, Multibody Dynamics, Durability, Optimization **FEA:** Introduction to CAE, Elements and Element Shapes General procedure to conduct FEA, Effective Utilization of FEA FEA Software's, Advantages and Limitations of FEA Key Assumptions in FEA **General working of FEA:** Stiffness matrix, Boundary conditions FEA through ANSYS Workbench Types of Engineering Analysis, Important terms and definitions, and Classification of materials	Theory/Demo	4.0 Hrs
2	Session 3	Exercises inclued for practice	Practicals	2.0 Hrs
3	Session 4&5	**INTRODUCTION TO ANSYS WORKBENCH** **ANSYS Workbench:** System requirements, Starting ANSYS Workbench, ANSYS Workbench GUI Working on a Project, Project Tab, Tool box in Project Tab, Configuring the Toolbox Analysis system, Component system, Custom system, Design exploration & External connection system Project Schematic, Adding system to Project Schematic **Components of the system:** Engineering Data cell, Geometry cell, Model cell, Setup cell, Solution cell and Results cell **System Editing:** Naming & Renaming systems, Moving, Deleting & Replacing systems, Duplicating systems Project Schematic links, Understanding the cell states, Menu bar, Units in ANSYS Workbench Working with files & Project	Theory/Demo	4.0 Hrs
4	Session 6	Exercises included for practice	Practicals	2.0 Hrs
5	Session 7&8	**CREATING MODEL FOR ANALYSIS USING SCDM** Introduction to Space Claim Modeling **Space Claim GUI:** Structure Tree, Selection Panel, Views, Status bar, Options Panel, Properties, Radial menu, Working with components **File operations:** New, Open, Save, Archive, Copy & Close Importing & Exporting objects, Locking and Unlocking objects, Selecting, Undo & Redo, Rectangle, Circle, Polygon,Tangent Arc, 3-Point Arc, Points **Sketch Editing:** Rounded corners, Trim Away, Offset curve, Dimensioning sketches **3D Modeling Tools:** Pull, Sweep, Drafting faces, Move, Combine, Split, Project tool, Datum Planes, Axis, Points,Origin, Patterns, Shell **Assembly Tools:** Tangent constraint, Align constraint, Orienting constraint Dimensions, Detaching & Checking Geometry	Theory/Demo	4.0 Hrs
6	Session 9	Exercises included for practice	Practicals	2.0 Hrs
7	Session 10&11	**PRE PROCESSING - MODEL PREPARATION IN SCDM** **Repair Menu:** Missing faces, Stitch, Gaps, Split edges, Extra edges, Duplicate faces, Merging faces, Small faces,Simplify tool, Inexact edges, Straightening faces, Relaxing surfaces, Tangency, **Prepare Menu:** Rounds, Faces, Bad faces, Extracting volume, Split by plane, Enclosures, Mid-surface, Spot, weld, Welds, Interference tool **Beams Group:** Creating Beams, Editing Beams, Orienting Beams, Connecting Beams, Splitting Beams, Beam, display, Moving beams	Theory/Demo	4.0 Hrs
8	Session 12	Exercises included for practice	Practicals	2.0 Hrs
9	Session 13&14	**ANSYS MECHANICAL APPLICATION** Introduction to ANSYS Mechanical application **GUI:** Main Menus, Standard Toolbar, Graphics Toolbar, Selection Utilities, Context Toolbar, Graphics Options,Toolbar, Edge Graphics Options Toolbar, Tree Outline, Status Symbols, Details View, Undefined or Invalid, Fields, Exposing Fields as Parameters, Geometry Window, Contextual Menu Options, Status Bar Contextual windows, Selecting geometry, Selecting nodes, Selecting Element & Element faces Pan, Rotate & Zoom Viewports, Section Planes, Wizards, Co-ordinate systems, Connections, Boundary conditions, Solver & Results	Theory/Demo	4.0 Hrs
10	Session 15	Exercises included for practice	Practicals	2.0 Hrs
		PRE PROCESSING - MESHING Meshing Overview, Meshing Implementation in ANSYS Workbench, Meshing Process in ANSYS		

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11	Session 16&17	**Types of Meshing:** Part/Body Level Meshing, Assembly Level Meshing, Meshing by Algorithm, Meshing by Element Shape **Mesh methods:** Patch Confirming, Patch independent, Tet Meshing, Hex Meshing, Hex/Prism/Tet Hybrid Meshing, Quad Meshing, Triangle Meshing **Mesh Controls:** Global Mesh Control & Local Mesh Control Trobuleshooting	Theory/Demo	4.0 Hrs
12	Session 18	Exercises included for practice	Practicals	2.0 Hrs
13	Session 19	Exercises included for practice	Practicals	2.0 Hrs
14	Session 20&21	**STATIC STRUCTURAL ANALYSIS** **Methods to Solve Engineering Problems:** Analytical method, Numerical method, & Experimental method Why do we carry out meshing & what is FEM? Accuracy while increasing nodes & elements **Various types of analysis:** Linear static analysis, Nonlinear analysis, Dynamic analysis, Buckling analysis, Thermal analysis, Fatigue analysis, Optimization, CFD analysis, Crash analysis, NVH analysis **Types of Failure theories:** Maximum shear stress theory, Maximum normal stress theory, Maximum distortion energy theory Structural Analysis in Ansys Workbench, points to remember Assembly analysis, Assembly analysis procedure **Contact types:** Bonded, No separation, Frictionless, Rough, Frictional	Theory/Demo	4.0 Hrs
15	Session 22	Exercises included for practice	Practicals	2.0 Hrs
16	Session 23	Exercises included for practice	Practicals	2.0 Hrs
17	Session 24&25	**MODAL ANALYSIS** Introduction to Modal Analysis **Dynamic Analysis:** Free Vibration, Forced Vibration Why Dynamic analysis Natural frequency analysis Different methods to solve free vibration analysis, Avoid Resonance Specifying analyzing settings	Theoy/Demo	4.0 Hrs
18	Session 26	Exercises included for practice	Practicals	2.0 Hrs
19	Session 27	**BUCKLING ANALYSIS** Introduction to Buckling Analysis **Types:** Eigen value buckling analysis, Nonlinear buckling analysis Points to remember	Theory/Demo	2.0 Hrs
20	Session 28	Exercises included for practice	Practicals	2.0 Hrs
21	Session 29&30	**FATIGUE ANALYSIS** Introduction to Fatigue Fatigue Analysis, S-N curve diagram Low cycle fatigue, High cycle fatigue Common decision for fatigue analysis, Why fatigue analysis Static, Dynamic & Fatigue Analysis comparison **Fatigue Results:** Fatigue life, Fatigue damage, Fatigue safety factor,	Theory/Demo	4.0 Hrs
22	Session 31	Exercises included for practice	Practicals	2.0 Hrs
23	Session 32	Exercises included for practice	Practicals	2.0 Hrs
24	Session 33&34	**THERMAL ANALYSIS** Introduction to Thermal Analysis Why Thermal analysis Conduction. Convection & Radiation Temperature **Types:** Steady state thermal analysis, Transient thermal analysis Preparing for analysis, Points to remember Thermal stress analysis **Important terms in thermal analysis:** Thermal gradient, Thermal flux, Bulk temperature, Film coefficient, Emissivity, Stephan Boltzmann constant, Thermal conductivity, Specific heat	Theory/Demo	4.0 Hrs
25	Session 35	Exercises included for practice	Practicals	2.0 Hrs
26	Session 36	Exercises included for practice	Practicals	2.0 Hrs
27	Session 37&38	**DESIGN OPTIMIZATION** Topology Optimization in ANSYS Workbench Geometrical parameters, Shape optimization Prerequisites and Requirements Design validation	Theory/Demo	4.0 Hrs
28	Session 39	Exercises included for practice	Practicals	2.0 Hrs
29	Session 40	ASSIGNMENT FOR EVALUATION	Practicals	2.0 Hrs